Zhengyang Hu

PhD Candidate · The University of Hong Kong
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Research areas: Tabular foundation model pretraining · Nonlinear correlation modelling · LLM reasoning

Education

The University of Hong Kong Aug 2023–Present
PhD in Electrical and Computer Engineering · GPA: 3.9 / 4.3
Advisors: Dr. Yanchao Yang and Prof. Kaibin Huang

University of Science and Technology of China · Major GPA: 3.87 / 4.3 Sep 2019–Jun 2023
BSc in Mathematics and Applied Mathematics · School of the Gifted Young

Research & Industry Experience

Minghong Investment — Machine Learning QR Intern Aug–Oct 2026

- Investigated whether dependence measures can serve as training losses. Using a fixed MLP architecture for five-day A-share return prediction, designed a post-training refitting strategy for dependence-based training, improving **mean cross-sectional test IC from 0.105 to 0.12** versus an MSE baseline (both 10-seed ensembles).

Huawei HPC Group (RAFT) — On-site Student Researcher Sep 2024–Jan 2026

- Developed representation-based diagnostics to monitor model training and validated them on NanoGPT, ViT and Pythia; the resulting research was submitted to ICLR.

ChengQi Asset Management — Quant Research Intern Aug–Nov 2022

- Researched alpha strategies using diverse data sources, with **five accepted** into the firm's alpha pool.

Selected Research & Publications

1. InfoNet: Neural Estimation of Mutual Information without Test-Time Optimization
Z. Hu et al. · **ICML 2024 (Oral)**
Developed a pretrained tabular model for mutual information estimation between random variables, achieving **1,000×** faster estimation and higher accuracy on low-dimensional benchmarks.

2. InfoAtlas: A Foundation Model for Zero-Shot Statistical Dependence Estimate
Z. Hu et al. · **ICML 2026**
Scaled InfoNet to high-dimensional dependence estimation through richer synthetic data and architecture design. Built data-generation and distributed pretraining pipelines for a **600M-parameter model** on **32 H200 GPUs**.

3. From Anisotropy to Anomaly: An Online Geometric Diagnostics Framework for Transformer Training
Z. Hu et al. · ICLR (under review)
Developed online diagnostics using statistical relationships among hidden representations to detect training anomalies, including overfitting, data anomalies and loss spikes during Transformer training.

4. DensityFM: Foundation-Style Pretraining for Density Estimation
Z. Hu et al. · ICLR (under review)
Pretrained a tabular foundation model for density estimation, outperforming all evaluated classical baselines on synthetic and real-world benchmarks. Applications include tail-risk estimation and outlier detection.

Additional Experience

Optiver Quant Focus — Three-day programme for PhD students. Sep 2026
Team placed 1st in the Quant Challenge; independently solved **5 of the 12** mathematics and brainteaser problems.

Susquehanna (SIG) PhD Brainteaser Battle Oct 2026
Team placed 2nd; independently solved the competition's hardest puzzle.

Tests: TOEFL iBT 105 · GRE 322 (Quant 170) | **Technical skills:** Python · PyTorch · pandas